

Manasa Rapuru

Scientific Software Engineer

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SUMMARY

Engineer with a bioinformatics background, translating scientific needs into data-driven applications and AI-enabled tools. I bridge biology, software engineering, and user-centered design to build practical systems for scientific work.

EDUCATION & CERTIFICATIONS

M.S. Bioinformatics: Boston University	Sept 2021 – May 2023
B.S. Microbiology & Chemistry Minor: San Jose State University	Aug 2016 – Dec 2020
Artificial Intelligence in Pharma and Biotech Certification: MIT Management Sloan School	Aug 2026
AWS Cloud Practitioner Certification	Expected Oct 2026
IBM AI Developer Certification: Coursera	Jan 2026
IBM Full Stack Software Developer Specialization: Coursera	Apr 2024
Cancer Biology Specialization: Coursera	Jul 2026

RELEVANT EXPERTISE

Communication & Product Skills: Requirements Gathering, Stakeholder Collaboration, User-Centered Development, Workflow Design, Product Iteration, Interdisciplinary Communication, Cross-Disciplinary Collaboration

AI & Machine Learning: Prompt Engineering, Claude Code, Machine Learning, RAG, LLM Applications, AI agents

Computing: Python, Linux, R (DESeq2, ggplot2), SQL, Bash, Git, Java, Perl, Snakemake

Data Science: Pandas, NumPy, SciPy, Scikit-learn, Matplotlib

Cloud & Infrastructure: AWS, HPC/SLURM, CI/CD

Domain Knowledge: Genomics, transcriptomics, CRISPR, single-cell RNA-seq, Bulk RNA-Seq, NGS, molecular assay design, qPCR/dPCR

Software Engineering: React, REST APIs, Fast API, Flask, Streamlit, HTML, CSS, Javascript, Docker, AWS, Application Development, SDLC

Bioinformatics Tools: Scanpy, Seurat, BLAST, dbSNP, COSMIC, SAMtools, BEDTools, STAR, GATK, FastQC, Trimmomatic, GV, Ensembl, NCBI

RELEVANT WORK

Podcast Cohost & Project Coordinator (unreleased yet) Aug 2026 - Present

- Exploring what becomes possible through interdisciplinary thinking in science. Examining how perspectives, ideas, and approaches from different fields can connect to foster new ways of thinking and problem solving.

UX Critic/Digital Story Teller Volunteer: California Academy of Life Sciences Aug 2026 - Present
Scientific Computing Department

- Using publicly available natural history archive databases to curate descriptions for academy's data
- Evaluating the user interface through interaction testing and reporting bugs

Member: US-Research Software Engineering Association Aug 2026 - Present
User Experience Working Group

- Learning how to gather and incorporate audience needs when developing scientific research software

Bioinformatics Scientist: Thermo Fisher Scientific

Apr 2024 – Nov 2025

qPCR/dPCR TaqMan Assays

- Worked within a revenue-generating commercial qPCR/dPCR assay design service that involved designing for SNP-level allele discrimination, expression quantification, and multiplexing requirements.
- Collaborated with cross-functional scientific, engineering, and product stakeholders to translate complex assay development workflows into deployable computational tools
- Developed AI-enabled and software applications to streamline data workflows, reduce turnaround time and improve cross-team efficiency

Bioinformatics Specialist: Massachusetts General Hospital

Mar 2021 – Aug 2023

Huntington's Transcriptomics/Genomics

- Developed a Python-based analytical pipeline (Pandas, NumPy, Matplotlib, Scikit-Learn) to model cell morphology metrics derived from IncuCyte imaging.
- Evaluated machine learning approaches to assess whether custom models could improve phenotypic classification accuracy in comparison to the instrument's default analysis software.
- Integrated RNA-seq, single-cell RNA-seq, and tandem-repeat genomic analyses to investigate disease-associated gene expression patterns and molecular signatures in Huntington's disease, implementing differential expression workflows using R, DESeq2, and ggplot2.
- Designed reproducible computational workflows for HPC environments, automating data processing in Bash and integrating custom and precompiled bioinformatics tools within shared Linux infrastructure.

Bioinformatics Master's Student: Boston University

Sept 2021 – May 2023

Bioinformatics

- Designed and developed an interactive scientific data application enabling researchers to explore postmortem brain transcriptomic datasets across brain regions and disease conditions.
- Built the application using Python, SQL, JavaScript, HTML, and CSS, implementing interactive data querying and visualization for large-scale transcriptomic datasets.
- Developed reproducible Snakemake-based workflows for RNA-seq and single-cell RNA-seq analysis in R and Linux environments, integrating upstream computational pipelines with the interactive analytical application.

Bioinformatics Intern: Sandia National Laboratories

Jun 2020 – Aug 2020

Bioinformatics Research / Phage Therapy

- Applied computational genomics to support phage therapy research, using phylogenetic and comparative analyses to investigate integrative genetic elements in bacterial genomes.
- Worked within HPC-based computational environments to execute and optimize bioinformatics workflows, diagnosing infrastructure constraints affecting memory, I/O, and job scheduling.
- Collaborated on the development and improvement of Python and Perl bioinformatics pipelines, contributing changes to open-source repositories through Git.

Undergraduate Bioinformatics Research Student: San Jose State University

Jan 2019 – Dec 2020

Phylogenetics & Protein Modelling

- Investigated evolutionary relationships and potential functional roles of *Candidatus Saccharibacteria* within the oral microbiome through phylogenetic and protein visualization analyses, translating complex biological datasets into interpretable scientific insights.